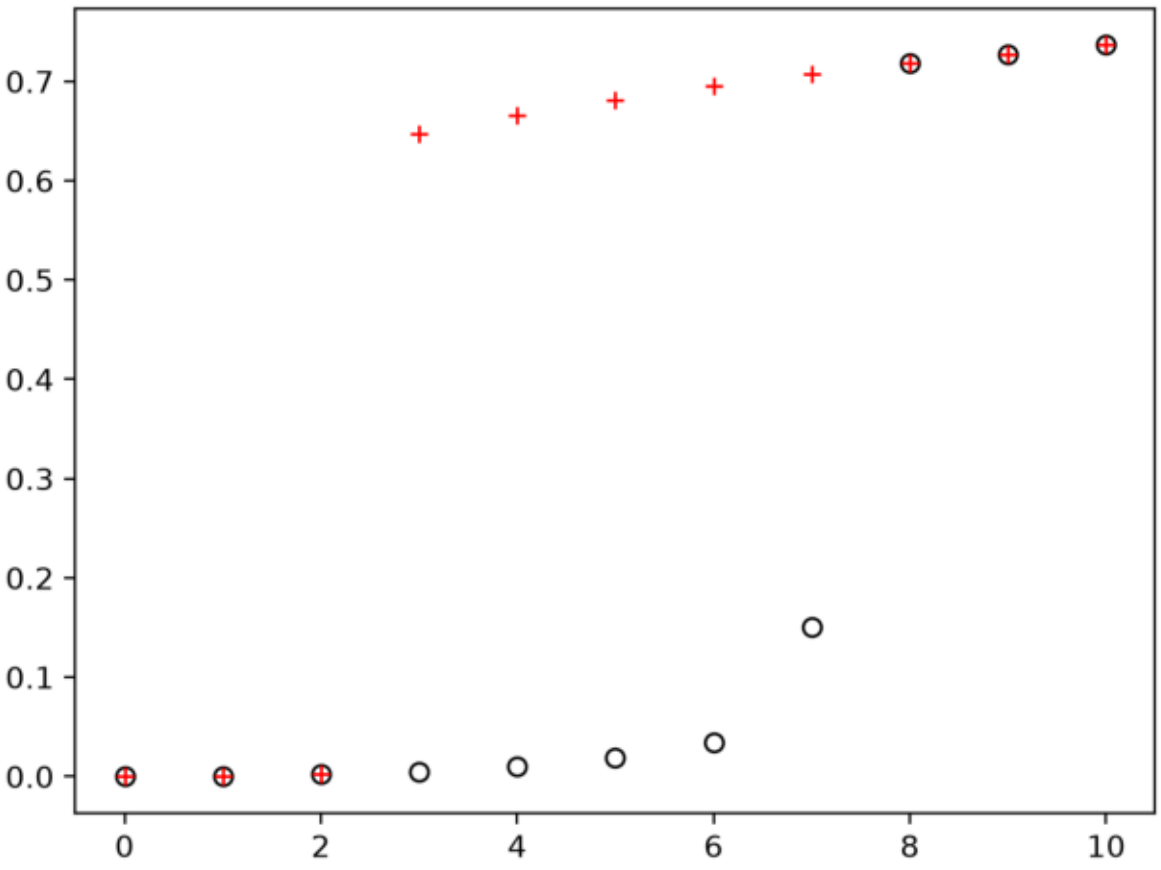
演習3

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.integrate import solve_ivp

def ozbudak_model( t , init , *param ):
    x = init[ 0 ]
    y = init[ 1 ]

    [ alpha , beta , rho , x_out ] = param

    dxdt = -x + alpha * x_out + beta * y
    dydt =  x**2 / ( rho + x**2 ) - y

    return [ dxdt , dydt ]

def ozbudak_dose_response( x_out , y_init , marker ):
    init = [ 0 , y_init ]
    t_span = [ 0 , 60 ]

    alpha = 0.1
    beta = 10
    rho = 25

    param = [ alpha , beta , rho , x_out ]

    solution = solve_ivp( ozbudak_model, t_span, init , method='RK45', args=param )

    LacY = solution.y[1]
    y_end = LacY[-1]

    plt.plot( x_out , y_end , marker , fillstyle='none')

    return y_end

##
## Main
##

y_init = 0
for x_out in range(11):
    y_init = ozbudak_dose_response( x_out , y_init , 'ko' )

for x_out in reversed( range(11) ):
    y_init = ozbudak_dose_response( x_out, y_init, 'r+')

plt.show()
```


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8
10